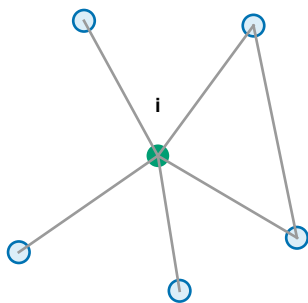
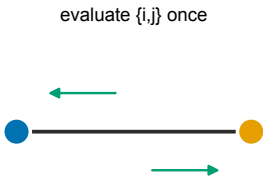


Conservative-compatible dynamic neural-SPH architecture

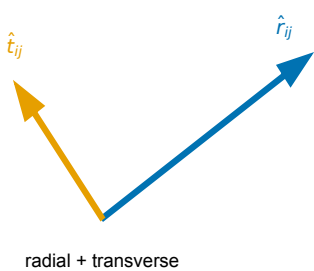
a Neighbor graph



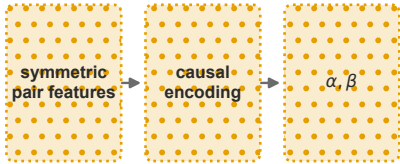
b One unordered edge



c Pair basis



d Coefficient head



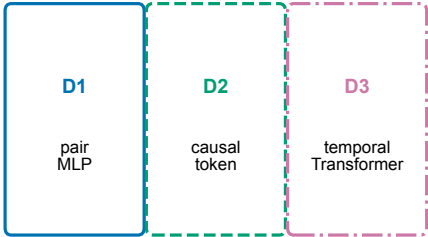
e Reciprocity

$$\mathbf{f}_{ij}^\theta = \alpha \hat{r}_{ij} + \beta \hat{t}_{ij}$$

$$\mathbf{f}_{ji}^\theta = -\mathbf{f}_{ij}^\theta$$

momentum-compatible by construction

f Comparison arms



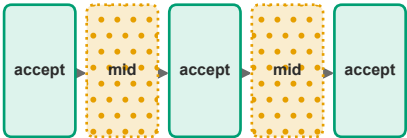
arms $\neq$ ranking

g Midpoint RK2



graph rebuilt at source evaluations

h Accepted history



history commit only after acceptance

i D0 identity

$$\Delta \mathbf{f}_{ij}^{D0} \equiv 0$$

$$\tau_{D0} = \tau_{SPH}$$

exact implementation identity